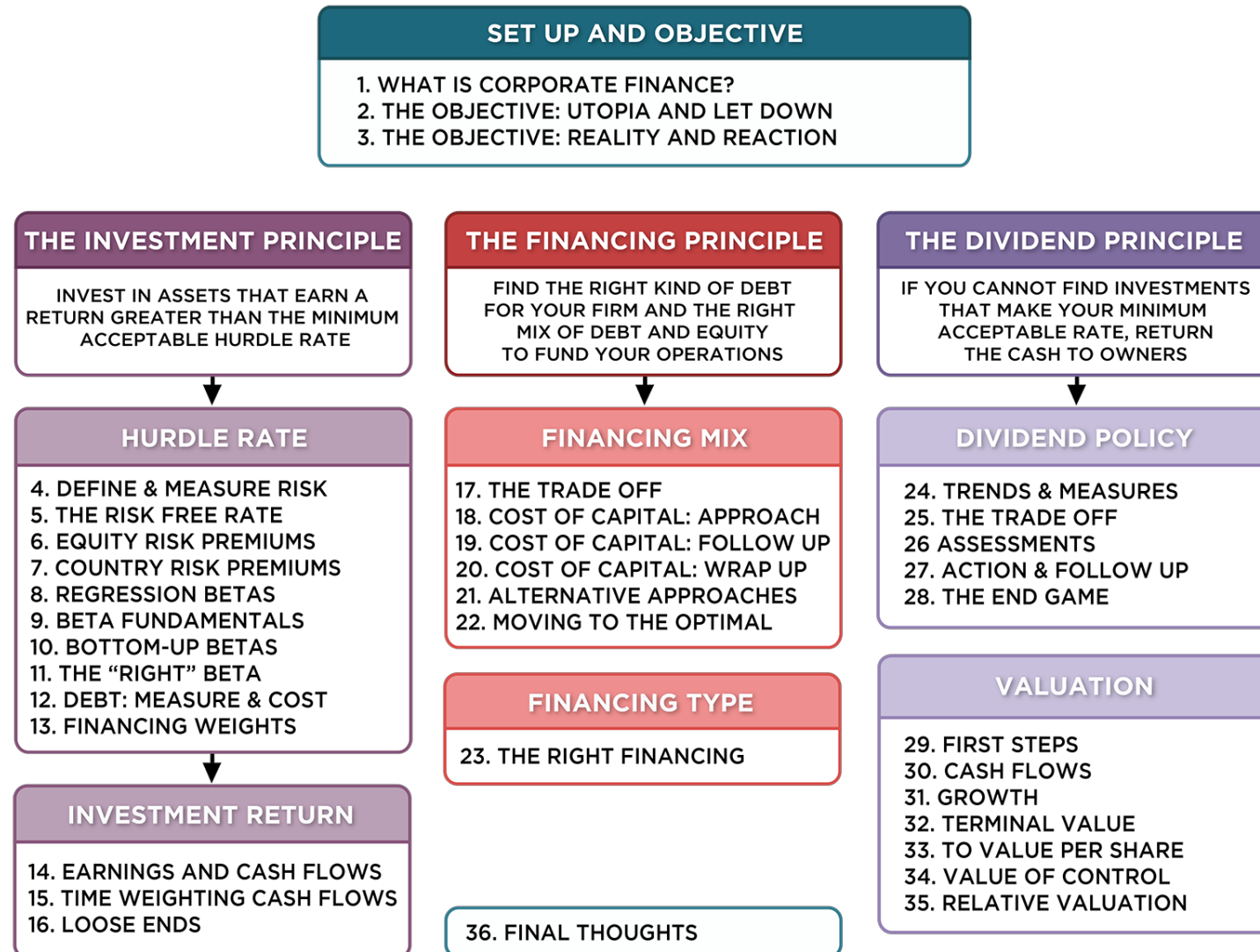


SESSION 13: FINANCING WEIGHTS

The minimum acceptable hurdle rate, at last..

SESSION 13: FINANCING WEIGHTS



WEIGHTS FOR COST OF CAPITAL CALCULATION

- The weights used in the cost of capital computation should be market values.
- There are three specious arguments used against market value
 - Book value is more reliable than market value because it is not as volatile: While it is true that book value does not change as much as market value, this is more a reflection of weakness than strength
 - Using book value rather than market value is a more conservative approach to estimating debt ratios: For most companies, using book values will yield a lower cost of capital than using market value weights.
 - Since accounting returns are computed based upon book value, consistency requires the use of book value in computing cost of capital: While it may seem consistent to use book values for both accounting return and cost of capital calculations, it does not make economic sense.
- In practical terms, estimating the market value of equity should be easy for a publicly traded firm, but some or all of the debt at most companies is not traded. As a consequence, most practitioners use the book value of debt as a proxy for the market value of debt.

DISNEY: FROM BOOK VALUE TO MARKET VALUE FOR INTEREST BEARING DEBT...

- In Disney's 2013 financial statements, the debt due over time was footnoted.

Time due	Amount due	Weight	Weight *Maturity
0.5	\$1,452	11.96%	0.06
2	\$1,300	10.71%	0.21
3	\$1,500	12.36%	0.37
4	\$2,650	21.83%	0.87
6	\$500	4.12%	0.25
8	\$1,362	11.22%	0.9
9	\$1,400	11.53%	1.04
19	\$500	4.12%	0.78
26	\$25	0.21%	0.05
28	\$950	7.83%	2.19
29	\$500	4.12%	1.19
	\$12,139		7.92

- Disney's total debt due, in book value terms, on the balance sheet is \$14,288 million and the total interest expense for the year was \$349 million. Using 3.75% as the pre-tax cost of debt:

- Estimated MV of Disney Debt =

$$349 \times \left[\frac{1 - \frac{1}{(1.0375)^{7.92}}}{.0375} \right] + \frac{14,288}{(1.0375)^{7.92}} = \$13,028 \text{ MILLION}$$

OPERATING LEASES AT DISNEY

- The “debt value” of operating leases is the present value of the lease payments, at a rate that reflects their risk, usually the pre-tax cost of debt.
- The pre-tax cost of debt at Disney is 3.75%.

YEAR	COMMITMENT	PRESENT VALUE @3.75%
1	\$507.00	\$488.67
2	\$422.00	\$392.05
3	\$342.00	\$306.24
4	\$272.00	\$234.76
5	\$217.00	\$180.52
6 - 10	\$356.00	\$1330.69
DEBT VALUE OF LEASES		\$2932.93

Disney reported \$1,784 million in commitments after year 5. Given that their average commitment over the first 5 years, we assumed 5 years @ \$356.8 million each.

- Debt outstanding at Disney = \$13,028 + \$ 2,933= \$15,961 million

APPLICATION TEST: ESTIMATING MARKET VALUE

- Estimate the
 - Market value of equity at your firm and Book Value of equity
 - Market value of debt and book value of debt (If you cannot find the average maturity of your debt, use 3 years): Remember to capitalize the value of operating leases and add them on to both the book value and the market value of debt.
- Estimate the
 - Weights for equity and debt based upon market value
 - Weights for equity and debt based upon book value

CURRENT COST OF CAPITAL: DISNEY

- Equity
 - Cost of Equity = Riskfree rate + Beta * Risk Premium
 $= 2.75\% + 1.0013 (5.76\%) = 8.52\%$
 - Market Value of Equity = \$121,878 million
 - Equity/(Debt+Equity) = 88.42%
- Debt
 - AFTER-TAX COST OF DEBT = (RISKFREE RATE + DEFAULT SPREAD) (1 - t)**
 $= (2.75\% + 1\%) (1 - .361) = 2.40\%$
 - Market Value of Debt = \$13,028+ \$2933 = \$ 15,961 million
 - Debt/(Debt +Equity) = 11.58%
- Cost of Capital = $8.52\%(.8842) + 2.40\%(.1158) = 7.81\%$

$121,878 / (121,878 + 15,961)$

DIVISIONAL COSTS OF CAPITAL: DISNEY AND VALE

DISNEY

BUSINESS	COST OF EQUITY	COST OF DEBT	MARGINAL TAX RATE	AFTER-TAX COST OF DEBT	DEBT RATIO	COST OF CAPITAL
MEDIA NETWORKS	9.07%	3.75%	36.10%	2.40%	9.12%	8.46%
PARKS & RESORTS	7.09%	3.75%	36.10%	2.40%	10.24%	6.61%
STUDIO ENTERTAINMENT	9.92%	3.75%	36.10%	2.40%	17.16%	8.63%
CONSUMER PRODUCTS	9.55%	3.75%	36.10%	2.40%	53.94%	5.59%
INTERACTIVE	11.65%	3.75%	36.10%	2.40%	29.11%	8.96%
DISNEY OPERATIONS	8.52%	3.75%	36.10%	2.40%	11.58%	7.81%

VALE

BUSINESS	COST OF EQUITY	AFTER-TAX COST OF DEBT	DEBT RATIO	COST OF CAPITAL IN US\$	COST OF CAPITAL IN R\$
METALS & MINING	11.35%	2.67%	35.48%	8.27%	15.70%
IRON ORE	11.13%	2.67%	35.48%	8.13%	15.55%
FERTILIZERS	12.70%	2.67%	35.48%	9.14%	16.63%
LOGISTICS	10.29%	2.67%	35.48%	7.59%	14.97%
VALE OPERATIONS	11.23%	2.67%	35.48%	8.20%	15.62%

COSTS OF CAPITAL: TATA MOTORS, BAIDU AND BOOKSCAPE

- To estimate the costs of capital for Tata Motors in Indian rupees:
 - Cost of capital= $14.49\% (1-.2928) + 6.50\% (.2928) = 12.15\%$
- For Baidu, we follow the same path to estimate a cost of equity in Chinese RMB:
 - Cost of capital = $12.91\% (1-.0523) + 3.45\% (.0523) = 12.42\%$
- For Bookscape, the cost of capital is different depending on whether you look at market or total beta:

	Cost of equity	Pre-tax Cost of debt	After-tax cost of debt	D/(D+E)	Cost of capital
Market Beta	7.46%	4.05%	2.43%	17.63%	6.57%
Total Beta	11.98%	4.05%	2.43%	17.63%	10.30%

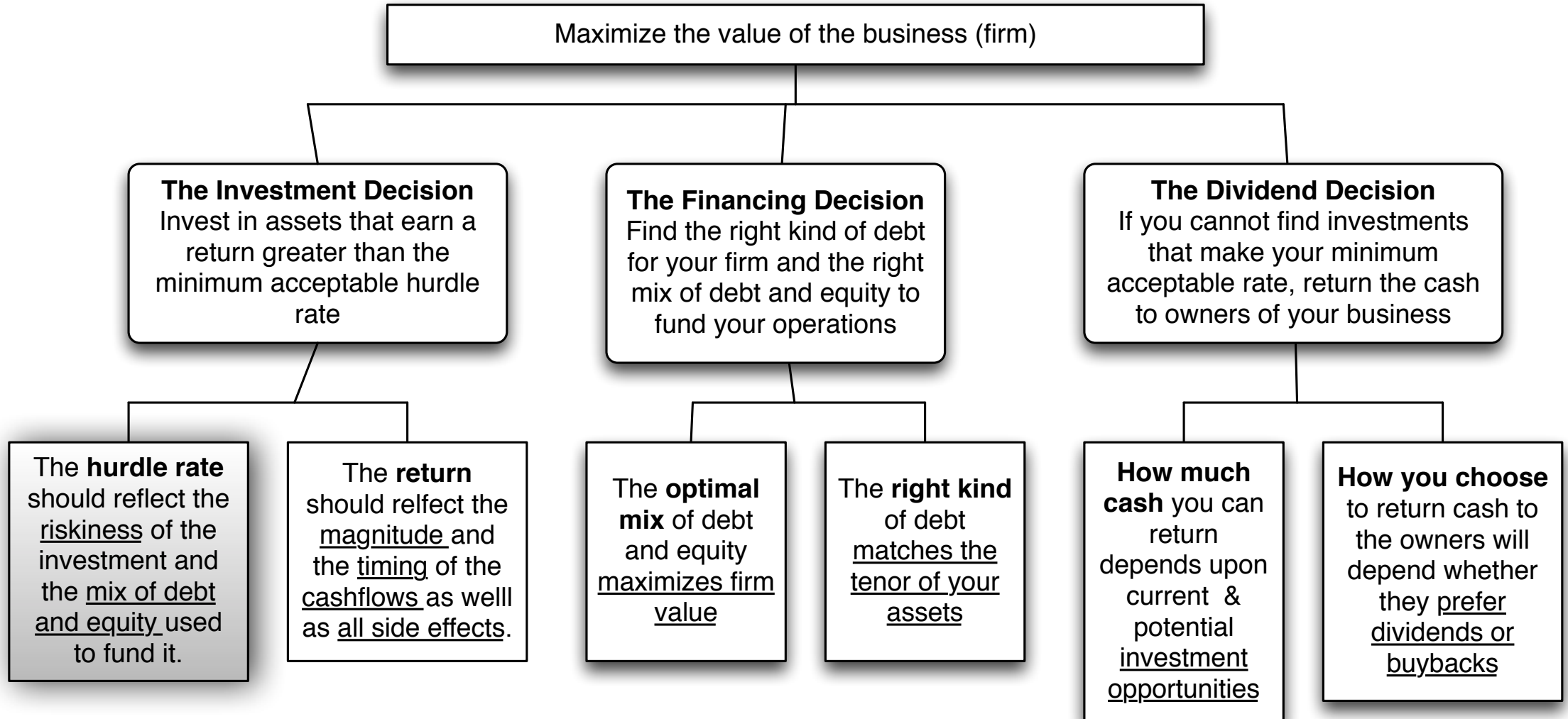
APPLICATION TEST: ESTIMATING COST OF CAPITAL

- Using the bottom-up unlevered beta that you computed for your firm, and the values of debt and equity you have estimated for your firm, estimate a bottom-up levered beta and cost of equity for your firm.
- Based upon the costs of equity and debt that you have estimated, and the weights for each, estimate the cost of capital for your firm.
- How different would your cost of capital have been, if you used book value weights?

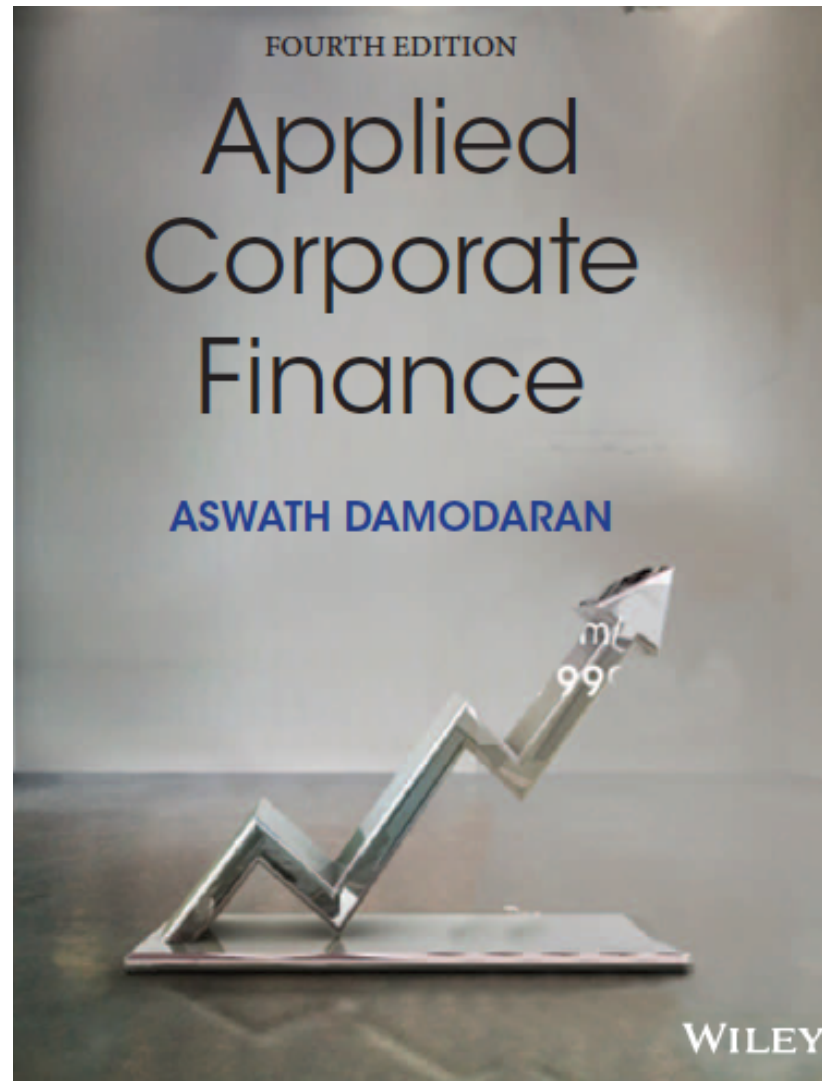
CHOOSING A HURDLE RATE

- Either the cost of equity or the cost of capital can be used as a hurdle rate, depending upon whether the returns measured are to equity investors or to all claimholders on the firm (capital)
- If returns are measured to equity investors, the appropriate hurdle rate is the cost of equity.
- If returns are measured to capital (or the firm), the appropriate hurdle rate is the cost of capital.

BACK TO FIRST PRINCIPLES



Task
Estimate a
cost of capital
for your
company (and
its businesses)



Optional:
Read
Chapter 4